

ARCTIC BAY CS, NU, Canada

WMO: 715920

Lat: 72.993N

Lon: 85.012W

Elev: 33

StdP: 14.68

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.1	-37.7	-49.0	0.3	-39.8	-46.4	0.4	-37.6	23.7	-6.5	21.0	-5.8	1.8	0	1.051

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	54.6	48.0	51.3	46.0	48.5	43.8	48.6	53.7	46.3	50.7	44.2	47.9	12.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
45.0	44.2	50.6	42.8	40.6	48.2	40.9	37.7	46.3	19.6	54.0	18.4	50.8	17.3	48.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	20.3	17.4	DB	-43.9	60.0	2.8	4.1	-45.9	62.9	-47.5	65.3	-49.0	67.6	-51.0	70.5	(4)
(5)			WB	-44.1	51.5	2.7	2.4	-46.0	53.3	-47.6	54.7	-49.1	56.0	-51.1	57.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-20.7	-25.3	-19.3	-1.9	19.9	36.1	41.8	41.8	32.1	17.9	-2.5	-13.4	(6)	
	DBStd	25.88	9.20	9.20	10.36	9.69	8.58	4.51	4.21	4.76	5.14	9.34	10.57	10.36	(7)	
	HDD50	14943	2190	2108	2149	1558	934	417	257	257	537	995	1575	1966	(8)	
	HDD65	20411	2655	2528	2614	2008	1399	867	719	718	987	1460	2025	2431	(9)	
	CDD50	7	0	0	0	0	0	0	3	4	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.5	3.8	3.4	4.4	5.0	6.9	8.6	8.2	7.9	8.6	9.3	6.6	4.9	(14)	
(15) Precipitation	PrecAvg	7.8	0.2	0.2	0.3	0.4	0.5	0.7	1.4	1.5	0.9	0.9	0.4	0.3	(15)	
	PrecMax	11.9	0.5	0.5	1.1	1.1	1.4	1.7	3.0	2.6	2.1	2.2	0.9	0.7	(16)	
	PrecMin	4.3	0.0	0.1	0.1	0.1	0.2	0.0	0.4	0.3	0.4	0.3	0.1	0.0	(17)	
	PrecStd	1.8	0.1	0.1	0.2	0.2	0.3	0.4	0.7	0.6	0.5	0.5	0.2	0.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	1.6	14.8	25.2	38.9	50.7	59.8	59.4	45.6	37.8	27.3	18.9	(19)	
		MCWB	16.1	0.6	13.1	23.2	35.1	43.5	50.0	52.1	41.5	34.6	24.3	17.1	(20)	
	2%	DB	2.5	-2.0	7.8	20.0	35.9	47.0	55.3	55.0	42.6	35.0	20.7	11.7	(21)	
		MCWB	1.2	-2.9	6.8	17.6	32.6	41.4	48.3	48.4	39.1	32.2	18.5	10.3	(22)	
	5%	DB	-3.1	-5.7	2.0	16.6	33.6	44.3	51.9	51.8	40.6	32.4	16.5	7.0	(23)	
		MCWB	-4.0	-6.5	0.8	14.6	30.8	39.7	46.5	46.5	37.6	30.0	14.7	6.0	(24)	
	10%	DB	-7.7	-11.3	-4.6	12.5	31.4	42.2	48.9	49.1	39.0	30.0	12.6	2.1	(25)	
		MCWB	-8.3	-11.8	-5.4	10.9	29.1	38.3	44.2	44.6	36.4	27.7	11.1	1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.0	0.5	13.2	23.1	35.5	44.6	51.2	53.5	42.0	35.0	24.8	17.4	(27)	
		MCDB	17.7	1.2	14.8	25.0	38.9	49.6	56.4	58.2	45.3	37.1	27.0	18.8	(28)	
	2%	WB	1.3	-2.8	6.8	17.7	32.9	41.9	48.9	49.2	39.5	32.5	18.5	10.4	(29)	
		MCDB	2.4	-1.9	7.8	19.7	35.4	45.9	54.8	53.7	42.1	34.9	20.5	11.5	(30)	
	5%	WB	-3.8	-6.5	1.0	14.6	31.0	40.0	46.6	46.8	37.8	30.3	14.8	5.9	(31)	
		MCDB	-3.0	-5.6	2.1	16.6	33.3	43.7	51.5	51.0	40.1	32.6	16.4	6.9	(32)	
	10%	WB	-8.3	-11.8	-5.4	10.9	29.2	38.4	44.5	44.9	36.6	27.9	11.2	1.1	(33)	
		MCDB	-7.7	-11.2	-4.7	12.5	31.4	41.7	48.7	48.4	38.8	29.9	12.7	2.0	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	10.2	11.8	14.2	10.4	7.8	9.6	7.8	5.4	7.2	8.6	9.8	(35)	
	5% DB	MCDBR	14.1	15.0	16.8	15.2	10.4	11.4	15.1	12.5	6.7	7.9	11.8	16.4	(36)	
		MCWBR	13.5	14.3	16.2	13.5	8.3	7.0	9.8	7.7	5.0	7.0	11.1	15.4	(37)	
	5% WB	MCDBR	14.0	15.0	16.5	14.9	10.1	11.0	14.4	11.7	6.5	7.3	11.6	16.0	(38)	
		MCWBR	13.5	14.3	16.0	13.4	8.3	7.2	9.9	7.8	5.0	6.7	10.9	15.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.170	0.239	0.293	0.314	0.312	0.328	0.311	0.263	0.180	N/A	N/A	(40)	
	taud		N/A	1.904	2.154	2.089	2.147	2.368	2.448	2.490	2.412	1.935	N/A	N/A	(41)	
	Ebn at Noon		0	134	231	255	267	275	265	255	229	117	0	0	(42)	
	Edh at Noon		0	9	20	33	36	30	27	23	16	8	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0	51	477	1307	1964	2001	1573	1018	409	97	0	0	(44)	
	RadStd		0	5	31	63	114	219	152	126	62	15	0	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page